

NACOGDOCHES, TX, USA

WMO: 722499

Lat: 31.578N

Lon: 94.709W

Elev: 355

StdP: 14.51

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 53948

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.9	27.6	11.9	10.2	32.1	17.6	13.5	37.2	22.0	52.6	19.5	52.8	2.6	350	0.389

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.9	99.1	75.8	96.8	76.0	93.3	75.9	79.4	89.9	78.5	89.2	78.0	88.7	6.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.1	143.0	82.7	75.5	135.2	81.9	75.1	133.4	81.8	43.1	89.6	42.2	89.9	41.6	88.3	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.4	16.3	13.9	17.7	100.8	4.3	3.7	14.6	103.5	12.1	105.7	9.7	107.8	6.6	110.5		
			16.6	81.0	3.9	1.2	13.8	81.9	11.5	82.5	9.3	83.2	6.4	84.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.1	48.0	52.2	58.5	65.6	73.6	79.8	82.4	82.7	76.9	66.7	56.8	49.3
	DBStd	14.56	10.05	10.21	9.29	7.67	6.02	4.03	3.32	3.78	5.48	7.94	9.18	9.65
	HDD50	450	161	86	31	2	0	0	0	0	0	1	36	133
	HDD65	2093	533	371	241	85	10	0	0	0	4	78	275	496
	CDD50	6332	99	149	294	472	732	893	1003	1012	806	519	240	113
	CDD65	2496	6	14	40	104	277	443	538	547	359	130	28	10
	CDH74	24977	16	65	250	832	2437	4494	5904	6214	3428	1146	162	29
	CDH80	11469	0	6	30	170	877	2095	3034	3327	1576	336	17	1
(14) Wind	WSAvg	5.7	6.3	6.6	6.9	7.1	6.3	5.4	4.5	4.5	4.2	4.9	5.4	5.8
(15) Precipitation	PrecAvg	49.6	4.6	4.4	4.3	4.2	4.3	4.3	3.0	3.5	3.7	4.3	4.6	5.0
	PrecMax	67.4	11.5	9.2	9.1	10.4	11.5	12.4	7.8	10.2	8.5	14.2	15.3	11.0
	PrecMin	25.7	0.7	0.2	1.4	1.1	0.1	0.7	0.5	0.0	1.8	0.0	0.2	1.2
	PrecStd	10.9	2.9	2.6	2.0	2.3	2.7	2.8	1.9	2.8	1.9	3.9	3.5	2.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.4	80.7	82.6	88.3	93.1	99.3	100.4	102.5	99.2	90.2	82.0	78.5
		MCWB	63.6	64.7	67.4	71.6	74.1	75.8	75.7	76.0	73.5	73.7	70.5	68.3
	2%	DB	72.6	75.2	80.7	83.9	90.3	95.3	98.8	99.2	93.3	87.6	78.9	73.0
		MCWB	62.4	64.2	66.4	70.0	74.2	76.3	76.1	76.0	74.1	73.2	67.7	65.7
	5%	DB	69.7	72.3	76.9	81.6	88.1	92.4	95.2	96.9	91.0	83.8	74.9	70.3
		MCWB	61.8	63.1	64.2	68.1	73.8	76.1	76.5	76.1	73.9	71.1	65.7	65.1
	10%	DB	65.7	69.5	73.1	79.3	84.4	90.3	92.7	93.4	89.5	81.1	72.3	66.2
		MCWB	60.4	62.0	63.0	66.5	72.6	76.2	76.6	76.1	73.5	69.2	65.1	61.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.4	69.2	70.8	74.4	78.3	80.4	80.5	80.4	79.2	77.7	73.1	70.3
		MCDB	72.8	74.5	78.5	82.9	87.6	89.2	90.0	91.4	88.3	85.2	79.4	74.2
	2%	WB	65.7	67.3	68.9	72.5	76.6	78.8	79.3	79.3	77.8	75.9	70.8	68.1
		MCDB	69.9	72.7	75.5	80.3	85.9	88.4	89.8	91.0	87.4	83.3	75.5	72.0
	5%	WB	63.8	65.5	67.3	71.1	75.4	77.9	78.4	78.4	76.8	74.2	68.9	66.1
		MCDB	67.0	70.7	73.5	78.5	84.0	88.0	89.2	90.0	86.0	80.4	73.4	69.5
	10%	WB	60.6	63.0	65.8	69.6	74.2	77.1	77.7	77.7	75.7	72.2	66.2	62.6
		MCDB	65.0	67.1	72.3	75.8	82.3	87.2	88.7	89.3	84.2	77.5	71.2	65.7
(35) Mean Daily Temperature Range	5% DB	MDBR	21.7	21.5	22.3	22.9	20.6	19.8	20.5	21.9	22.2	23.3	22.4	21.7
		MCDBR	24.8	23.9	23.6	23.3	22.0	22.5	24.1	25.4	25.3	24.1	23.9	23.3
		MCWBR	16.1	14.7	12.9	11.7	9.2	7.9	7.0	7.1	9.4	11.8	14.6	16.7
	5% WB	MCDBR	18.5	19.0	19.1	18.3	18.9	18.8	20.3	21.4	20.1	19.0	19.7	18.9
		MCWBR	14.4	13.7	12.6	10.5	9.2	7.9	7.4	7.1	8.3	10.8	14.1	15.4
(40) Clear-Sky Solar Irradiance	taub		0.325	0.344	0.362	0.407	0.443	0.462	0.477	0.472	0.430	0.368	0.357	0.332
	taud		2.522	2.477	2.427	2.307	2.250	2.244	2.234	2.243	2.319	2.474	2.455	2.511
	Ebn at Noon		286	291	292	281	270	264	259	259	266	278	271	277
	Edh at Noon		27	31	35	41	44	44	44	44	39	31	29	26
(44) All-Sky Solar Radiation	RadAvg		882	984	1375	1700	1857	1982	2002	1891	1609	1319	967	765
	RadStd		80	171	129	109	137	148	146	142	148	181	111	96

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+153

Nomenclature: See separate page